

RetroDiff: Training Objective Derivation

Let x denote a clean data sample and x_t its noised version at diffusion step $t \in \{1, \dots, T\}$. The forward process $q(x_{1:T}|x)$ gradually corrupts x ; the model p_θ learns to reverse it. Applying Jensen’s inequality to the intractable log-likelihood yields the *evidence lower bound* (ELBO):

$$\log p_\theta(x) \geq \mathbb{E}_{q(x_{1:T}|x)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x)} \right] = -\mathcal{L}_{\text{ELBO}} \quad (1)$$

Because the inequality is tight only when the variational posterior matches the true posterior, maximising the right-hand side of (1) is equivalent to minimising $\mathcal{L}_{\text{ELBO}}$. Expanding the joint $p_\theta(x_{0:T})$ and applying the Markov structure of the forward process decomposes the bound into a reconstruction term and a sum of per-step KL divergences:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x_0|x_1)]}_{\text{reconstruction}} + \sum_{t=2}^T \underbrace{\text{KL}(q(x_{t-1}|x_t, x) \parallel p_\theta(x_{t-1}|x_t))}_{\text{step-}t \text{ denoising}} \quad (2)$$

Each KL term in (2) measures how well the learned reverse step $p_\theta(x_{t-1}|x_t)$ approximates the tractable posterior $q(x_{t-1}|x_t, x)$; both are Gaussian under the standard linear noise schedule, so each term admits a closed form.

RetroDiff extension. The key contribution of RetroDiff is to condition the entire objective on a set of retrieved exemplars $R(x)$ drawn from the training corpus at inference time. Concretely, the per-sample ELBO is evaluated with $R(x)$ appended to the model’s context, and the final training objective averages over the data distribution:

$$\mathcal{L}_{\text{RetroDiff}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\mathcal{L}_{\text{ELBO}}(x \mid R(x))] \quad (3)$$

Equation (3) reduces to the standard diffusion objective when $R(x) = \emptyset$, making RetroDiff a strict generalisation. The retrieval set $R(x)$ is frozen during the backward pass; only θ is updated.